

MATCH 3 GAME DOC

Variables

1. tileColourArray: stores the tile colour index or property of that tile index
2. tileAnimArray: stores the animation index of each tile index

Functions

1. ACustomGameMode(): Constructor – Game Mode
2. deleteTile(int tileIndex): Inputs the tileIndex and deletes it from the grid
3. tileClicked(int tileIndex): On tile click, the possibilities of events to happen
4. ifTileMove(int tileIndex): If tile moves from the index, downward movement effect is shown
5. ifGridVoid(int tileIndex): If there are empty spaces in the grid after downward movement, rest of grid is filled i.e., Regeneration
6. shuffleGrid(): shuffles the game grid

Work Ahead of Us:

~~1. Randomness of Generation~~

2. Systematic Code of the project

3. New Icons setup along with newly generated background

4. Layering of the grid

5. Animation Integration

6. Swap Through Logic

7. Special Gem Combo Logic

8. Menu Page

Unreal – Match 3 Tutorial

https://www.youtube.com/playlist?list=PLZlv_N0_O1gZ-Qe3XN0R88UAPfa79kxJL

Meeting (26th Sept):

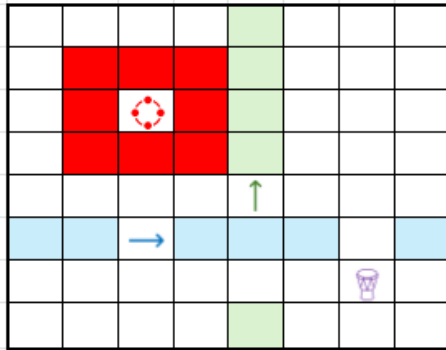
Refer to the special Case excel - [SpecialCases.xlsx \(sharepoint.com\)](#)

4 Bomb(watermelon)

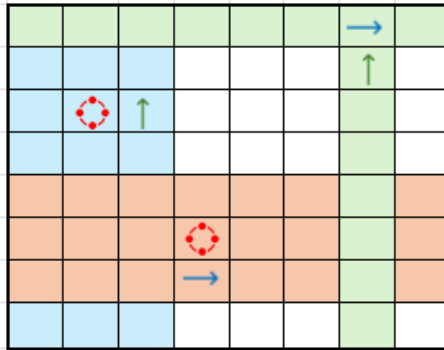
5 sugarcane(horizontal/vertical)

6+ Pineapple: removes a type of fruit

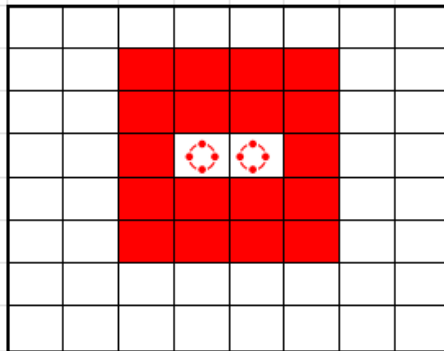
Case 1



Case 2



Case 3



- 4 Watermelon (bomb) 
 5 Sugercane (Hor/Ver)  

6+ Pineapple 

Case 4



List Of Classes

1. ActorClass – **BaseTile**

1.1 PaperSpriteActor* (denote type of Fruit/gem/special powerup)

1.2 OnClick Function

1.3 SpriteSpecialCondition (Grassed, SemiGrassed, Ice/semi, Wooden)

```
BaseTile(SpecialConditionSprite*) {  
    sprite = ();  
}
```

//Manager class between tile and gameMode! game instance class what it does

2. PaperSpriteActor -

2.1 OnClick()

Enum SpriteType { Fruit, Gem }

Enum FruitType {A, B, G, W}

Enum GemType {Bomb, Sugarcane, Pineapple } OR

Class SpriteType{

Enum FruitType {A, B, G, W }

Enum GemType {BOMB, SUGARCANE, WATERMELON}

}

SpriteType :: FruitType

SpriteType :: GemType

2.1 Fruit Classes

Apple

SpriteType::FruitType = Apple;

Sprite = Apple_Sprite

OnClick:: Clustering

Banana

Grape

Watermelon

2.2 Gem Classes

Bomb

```
SpriteType::GemType = bomb;  
Sprite // Deletion Array or storing covered area
```

Structure of Code:

GridActor -> TileActor -> PaperSprite Class ->

GridActor:

1. NxN tiles spawned: TArray<TileActor*>
2. Mapping using double way map or array
- 3.

TileActor:

1. TileActor(paperSprite) for inbuild obstacles
2. PaperSprite Class
- 3.

PaperSpriteClass:

1. Sprite type class
 - 1.1 Fruit or Gem Enum
2. Sprite
3. Fruit and Gem classes
 - 3.1 OnClick Logic
 - 3.1.1 Clustering logic with their animation
 - 3.1.2 Gem Blast logic with their animation
 - 3.1.3 Downward Movement w animation
 - 3.1.4 Regeneration with animation